

Upstream context

Session-level
raw trajectory / GPX

DEM / terrain-
elevation context

Session-level
HR stream

Core analytical workflow

Trajectory parsing +
elevation enrichment

Downhill-run
segmentation

HR anchoring and
run-level slicing

Carving-focused
staged labeling

Run-level
table

Internal run-level
summary

Mechanical run-level
summary

Within-dataset standardization

z_internal

z_mech

CL(α)

Alpha-sweep
diagnostics

**Comparison
descriptors**

Top-k
consistency

Phase-wise
contrast

Robustness extension

Perturb

Recompute

Compare

Summarize